

Jaden (Minjae) Moon

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Research interests: trustworthy machine learning, selective prediction and uncertainty, multimodal and health time-series modeling

Education

Dartmouth College, B.A. in Computer Science — Hanover, NH

Expected June 2027

GPA: 3.92 / 4.00

Relevant Coursework: Deep Learning; Deep Learning Robustness; Machine Learning; Artificial Intelligence; AI Agents; Software Design & Implementation; Linear Algebra; Differential Equations

Publications & Manuscripts

Moon, J., Pillai, A., & Campbell, A. T. (2026). *When Does Quality-Aware Multimodal Fusion Matter? A Leakage-Safe Diagnostic for Decision-Level Dependence*. To appear in *Proceedings of INTERSPEECH 2026*.

Accepted

Moon, J., Wu, Y., Pillai, A., & Campbell, A. T. (2026). *ReliaGate: Reliability Routing for Low-Stakes Wearable Stress Prediction*. WellComp 2026, Workshop at UbiComp/ISWC 2026.

Under review

Research Experience

Mayo Clinic Comprehensive Cancer Center

Jun 2026 – Present

Summer Undergraduate Research Fellow | Advisor: Wei Liu, Ph.D.

- Developed leakage-audited **selective-prediction** methods for quality assurance of AI-generated head-and-neck organ contours using 3D CT, contour geometry, segmentation uncertainty, and fixed native-plan dose context. Built and locked a selective-review pipeline that designated **59.4% of 3,812 held-out contour instances** as low risk at **99.91% NPV, 97.47% sensitivity**, and **0.936 AUROC**; froze the model and operating threshold before test evaluation.
- Designed a patient-disjoint dense spatial-error framework spanning **752 cases, 37,228 contour rows, and 5 folds**; completed verified preprocessing for three pilot folds using nnU-Net and SLURM, with patient-exclusion and storage guards, SHA-256 provenance, create-once artifacts, and a **134-test** validation suite.

HealthX Lab, Dartmouth College

Sep 2025 – Present

James O. Freedman Presidential Scholar | Advisor: Prof. Andrew T. Campbell

- ReliaGate (WellComp 2026, under review):** Proposed fixed-label reliability routing for wearable stress systems using confidence, signal quality, cross-modal agreement, and train-fitted representation geometry to surface or withhold unchanged predictions.
- Evaluated ReliaGate across **WESAD, UBFC-Phys, EmpaticaE4Stress, and PhysioNetE4** using subject-disjoint folds and paired held-out-subject intervals; on UBFC-Phys, increased coverage by **10.1 percentage points** while reducing accepted risk by **5.6 percentage points**
- Uncovered a pooled-versus-subject reliability trade-off: favorable aggregate routing outcomes coexisted with higher median subject-level risk, motivating subject-level evaluation of output availability and reliability.
- Quality-aware fusion diagnostic (INTER_SPEECH 2026, accepted):** Developed a leakage-safe post-hoc test that freezes unimodal experts and fusion parameters while breaking quality-instance alignment at inference time.
- Built controlled pipelines on **StressID** and **CMU-MOSEI** across 5 seeds $\times$ 5 disjoint folds; native quality signals produced near-zero Clean-Broken gaps, while positive controls verified diagnostic sensitivity when reliability tracked expert correctness.

AIM HIGH Lab, Dartmouth College

Feb 2025 – Aug 2025

Waterhouse Research Award | Advisor: Prof. Nicholas C. Jacobson

- Fine-tuned **LLaMA-3.1-8B-Instruct** with LoRA/QLoRA for therapeutic-dialogue generation under constrained compute.
- Engineered an evaluation pipeline for **1,753 generated responses**, combining semantic metrics with structured error analysis.
- Reduced training cost by **35%** using 4-bit quantization, enabling broader experimental sweeps under fixed compute budgets.
- Designed prompting and inference workflows to improve contextual grounding and reduce off-target generations in mental-health dialogue synthesis.

Honors, Awards & Research Grants

Mayo Clinic Summer Undergraduate Research Fellowship (SURF) – \$6,000	2026
Passport AI Grant, GoPassport Education (External grant for Mayo Clinic SURF) – \$2,000	2026
Best Class Project Citation – Deep Learning Robustness (COSC 89.31)	Winter 2026
James O. Freedman Presidential Scholar – \$3,400	2025
Waterhouse Research Award (Summer Research Grant) – \$6,700	2025
Dean’s Honor List	2023–2026

Technical Skills

Machine Learning: PyTorch, scikit-learn, multimodal and time-series modeling, selective prediction, uncertainty calibration, statistical evaluation
Programming & Infrastructure: Python, NumPy, pandas, SciPy, Git, Linux, SLURM, Google Colab, AWS, Flask/FastAPI
Biomedical Data: PPG/BVP, EDA, ECG, and temperature-signal processing; DICOM, NIfTI, NeuroKit2, BioSPPy, OpenSMILE

Selected Technical Project

Local Mental Health Finder Live Demo | Code
• Built and evaluated a full-stack therapist-discovery system using symptom-based search and learning-based provider ranking under sparse, noisy inputs.

Leadership & Service

Republic of Korea Marine Corps (ROKMC) Sep 2021 – Mar 2023
Communications Specialist & Interpreter (Squad Leader) – Hwaseong, South Korea
• Led a **20-person unit** as Squad Leader and initiated “Weekly Hour,” a peer-support program designed to identify unspoken distress in high-stigma environments.
• Managed base networking infrastructure at headquarters and resolved **150+ IT support cases monthly** to maintain operational continuity; provided technical interpretation during joint U.S.–ROK exercises.

Dartmouth Tech Consulting Group Sep 2024 – Present
• Led web development and digital strategy projects for campus organizations, delivering production-grade systems under client and timeline constraints.

Korean Student Association Sep 2023 – Present
• Organized cultural exchange events and mentorship programs supporting international students.

Languages

Korean (Native), English (Full Professional Proficiency)